

Exception Handling in Spring Boot

❏ What is Exception Handling?

- Handling **runtime errors** gracefully
 - Prevents app crash
 - Returns **proper HTTP response** instead of stack trace
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❏ Why Exception Handling is needed?

- Clean error responses (JSON)
 - Centralized error logic
 - Better debugging
 - Professional APIs
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❏ Basic Ways of Exception Handling

Way	Where	Use
try-catch	Method	Small logic
throws	Method signature	Propagate error
@ExceptionHandler	Controller	Controller-level
@ControllerAdvice	Global	Best practice
@RestControllerAdvice	Global	Best

1. ExceptionHandler (Controller Level)

```
@ExceptionHandler(RuntimeException.class)

public ResponseEntity<String> handle(RuntimeException ex)

{

    return ResponseEntity.badRequest().body(ex.getMessage());

}
```

✓ Handles exception in **same controller**

2. @ControllerAdvice (GLOBAL – BEST)

@RestControllerAdvice

```
public class GlobalExceptionHandler
```

```
{  
    @ExceptionHandler(RuntimeException.class)  
    public ResponseEntity<String> handle(RuntimeException ex)  
    {  
        return ResponseEntity.status(404).body(ex.getMessage());  
    }  
}
```

✓ Handles exceptions for **all controllers**

3. Custom Exception (VERY IMPORTANT)

```
public class ResourceNotFoundException extends RuntimeException
```

```
{  
    public ResourceNotFoundException(String msg)  
    {  
        super(msg);  
    }  
}
```

Use:

```
throw new ResourceNotFoundException("Employee not found");
```

Common HTTP Status Codes

Code	Meaning
400	Bad Request
404	Not Found
500	Server Error
401	Unauthorized
403	Forbidden

11 Standard Error Response (JSON)

```
{  
  "timestamp": "2026-01-10",  
  "status": 404,  
  "error": "Not Found",  
  "message": "Employee not found",  
  "path": "/employees/1"  
}
```

Exception Handling Flow (Steps)

1. Error occurs
 2. Exception thrown
 3. Spring searches handler
 4. @ExceptionHandler matches
 5. Response sent to client
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11 Best Practices (RULES)

- ✓ Use **@ControllerAdvice**
- ✓ Create **custom exceptions**
- ✓ Never expose stack trace
- ✓ Use proper HTTP status
- ✓ Keep controllers clean